METRIC

MIL-C-22325B(SH)
25 July 1988
SUPERSEDING
DOD-C-22325A(SH)
23 December 1977
(See 6.6)

MILITARY SPECIFICATION

# COLORS, TINTING, FOR INTERIOR NONFLAMING (DRY) PAINTS (METRIC)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers tinting colors to be used by mixing with enamel, nonflaming (dry), chlorinated alkyd resin, soft white semigloss formula no. 124 of DOD-E-24607 to produce interior finish paints for living spaces on ships. Products covered by this specification are suitable for use where air pollution regulations apply.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8010
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SPECIFICATIONS

FEDERAL

TT-T-291 - Thinner, Paint, Mineral Spirits, Regular and Odorless.
PPP-P-1892 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.

MILITARY

DOD-R-21417 - Resin, Chlorinated Alkyd, Solution. (Metric)
DOD-E-24607 - Enamel, Interior, Nonflaming (Dry), Chlorinated Alkyd Resin, Semigloss. (Metric)

STANDARDS

FEDERAL

FED-STD-141 - Paint, Varnish, Lacquer and Related Materials: Methods of Inspection, Sampling and Testing.
FED-STD-313 - Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities.
FED-STD-595 - Colors.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 209 - Standard Specification for Lampblack Pigment.
D 261 - Standard Sepcification for Iron Blue Pigment.
D 562 - Standard Test Method for Consistency of Paints Using the Stormer Viscometer. (DoD adopted)
D 656 - Standard Specification for Pure Toluidine Red Toner.
D 1210 - Standard Test Method for Fineness of Dispersion of Pigment-Vehicle Systems. (DoD adopted)
D 1296 - Standard Test Method for Odor of Volatile Solvents and Diluents. (DoD adopted)
D 1364 - Standard Test Method for Water in Volatile Solvents (Fischer Reagent Titration Method). (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)



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(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Toxicity. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the Naval Medical Command (NAVMEDCOM) which will act as advisor to the contracting activity (see 4.5).

3.2 Formula. The tinting colors shall consist of ingredients conforming to the applicable specifications in the proportions shown in table I, except that the pigment proportions may be varied slightly to conform to the color requirement. When specified in the contract or order, a certificate of compliance shall be prepared (see 6.2.2).



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TABLE I. Tinting color formulas.

| Ingredients | Kilograms (kg) (pounds) per 100 gallons 1/ | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Rose wood | Beach sand | Yellow gray | Green gray | Pearl gray | Sun glow | Clipper blue | Pastel blue | Pastel green | Blkhd gray |
| Yellow | 5/ | 5/ | 5/ | 5/ | 5/ | 5/ | 5/ | | 5/ | 5/ |
| Toluidine red (ASTM D 656) | 28.182 (62) | 15.909 (35) | 6.0 (13.2) | 2.636 5.8 | ---- | 4.091 (9) | ---- | 1.091 (2.4) | ---- | 7.273 (16) |
| Lampblack (ASTM D 209) | 1.182 (2.6) | 1.409 (3.1) | 4.0 (8.8) | 8.682 (19.1) | 9.545 (21) | 0.182 (0.4) | 5.409 (11.9) | 4.727 (10.4) | ---- | 3.182 (7) |
| Iron blue (ASTM D 261) | ---- | ---- | ---- | ---- | ---- | ---- | 5.909 (13.0) | 6.591 (14.5) | 5.909 (13.0) | ---- |
| Magnesium silicate 2/ | 52.273 (115) | 81.818 (180) | 110.455 (243) | 126.818 (279) | 136.364 (300) | 28.182 (62) | 123.182 (271) | 139.091 (306) | 136.364 (300) | 81.818 (180) |
| Chlorinated alkyd resin (DOD-R-21417) | 181.818 (400) | 168.182 (370) | 163.636 (360) | 156.818 (345) | 156.818 (345) | 197.727 (435) | 163.636 (360) | 156.818 (345) | 156.818 (345) | 172.727 (380) |
| Petroleum spirits (type I of TT-T-291) | 107.273 (236) | 119.091 (262) | 124.091 (273) | 130.0 (286) | 134.091 (295) | 95.455 (210) | 124.091 (273) | 134.091 (295) | 134.091 (295) | 118.182 (260) |
| Driers | 4/ | 4/ | 4/ | 4/ | 4/ | 4/ | 4/ | 4/ | 4/ | 4/ |
| Antiskinning/ antisettling agent | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ | 3/ |

1/ The formula is given slightly in excess of 100 gallons to allow for normal manufacturing loss.
2/ The magnesium silicate shall be a micronized grade equal to "S.F. magnesium silicate" marketed by Whittaker-Clark and Daniels.
3/ The amounts of antiskinning agent and antisettling agent may be adjusted to meet the requirements of the finished enamel. The agents shall be effective for their purpose in small amounts in the order indicated.
4/ The type and amount of driers shall be selected by the contractor. When used to tint formula 124 of DOD-E-24607, the dry time requirements specified in DOD-E-24607 shall be met.
5/ If necessary, the enamel shall be tinted with a lead-free, asbestos-free, color-and heat-stable yellow pigment to obtain the specified FED-STD-595 color.





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3.2.1 Recovered materials. Unless otherwise specified herein, all material incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3 Manufacture. The component raw materials shall be mixed and ground as required to produce a tinting color which is uniform, easily dispersed in the white base (formula no. 124), free from grit, free from color separation, and in complete conformance to the requirements specified herein.

3.4 Quantitative requirements. The tinting colors, prepared as specified (see table I), shall conform to table II (see 4.4).

TABLE II. Quantitative requirements.

| Characteristic | Minimum | Maximum |
|---|---|---|
| Water, percent by weight of tinting color | ---- | 0.5 |
| Coarse particles and skins (as residue retained on standard no. 325 sieve) percent by weight of tinting color | ---- | 0.5 |
| Consistency, Krebs-Stormer shearing rate 200 r/min | | |
| grams | 125 | 175 |
| equivalent Krebs units | 67 | 77 |
| Fineness of grind | 6 | ---- |
| Lead content (percent) | ---- | 0.06 |

3.4.1 Volatile portion. The volatile portion of the tinting colors shall not exceed 340 grams of solvent per liter of tinting color.

3.5 Qualitative requirements.

3.5.1 Odor. The odor shall be characteristic of the volatile portion permitted in 3.4.1.

3.5.2 Color. The dry film of the paint shall acceptably match the applicable standard color of FED-STD-595 as specified (see table III and 4.4.2.2).

TABLE III. Color numbers of paints made from the tinting pigments.

| Color | Rose wood | Beach sand | Yellow gray | Green gray | Pearl gray | Sun glow | Clipper blue | Pastel blue | Pastel green | Bulkhead gray |
|---|---|---|---|---|---|---|---|---|---|---|
| FED-STD-595 color no. | 22519 | 22563 | 26400 | 26496 | 26493 | 23697 | 24516 | 25526 | 24585 | 26307 |



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3.5.3 Compatibility. There shall be no evidence of incompatibility of any the ingredients of the tinting color or of the resultant paint when mixed as specified in 4.4.2.2.

3.5.4 Condition in container. For at least 1 year after date of manufacture, the tinting color, prepared as specified (see table I), shall be such that it may be readily broken up with a paddle to a smooth, uniform consistency, shall not exceed 90 Krebs units in viscosity, and shall not liver, curdle, gel, or show any other objectionable properties.

3.6 Material safety data sheet. The contracting activity shall be provided a material safety data sheet (MSDS) at the time of contract award. The MSDS is Form OSHA-20, found in and part of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification (see 6.4).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Quality conformance inspection. Quality conformance inspection shall be provided in accordance with method 1031 of FED-STD-141 as hereinafter supplemented. Submission of forms in accordance with method 1031 shall include exact formula and shall cover complete tests other than the 1-year condition in container test. The condition in container requirement shall apply for 1 year after manufacture regardless of other testing or prior acceptance of the material. Inspection shall particularly determine precision of fill.

4.2.1 Ingredient materials. When requested by the controlling authority, 1 pint of the ingredients in the formula in table I shall be provided for test purposes.

4.3 Acceptance. Acceptance of the tinting color shall be based upon conformance of the tinting color to the requirements of this specification. Failure of any test shall be cause for rejection of the lot.



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4.4 Test procedures.

4.4.1 Ingredient materials. Ingredient materials submitted shall be tested to determine compliance with the applicable specifications listed in table I.

4.4.2 Finished tinting color.

4.4.2.1 Tests. The following tests shall be conducted in accordance with the methods specified in table IV:

TABLE IV. Test methods.

| Tests | Applicable method in FED-STD-141 | Applicable ASTM test method |
|---|---|---|
| Water | ------ | D 1364 |
| Coarse particles and skins | 4092.1 | ------ |
| Consistency | ------ | D 562 |
| Fineness of grind | ------ | D 1210 |
| Odor | ------ | D 1296 |

4.4.2.2 Color. The color shall be determined as follows. Add the entire contents of the 1/2-pint can of tinting color to 1 gallon of soft white chlorinated alkyd resin paint (formula no. 124) and mix thoroughly. (A suggested procedure is as follows. After draining the 1/2-pint can of tinting color into the soft white paint with constant stirring, partly refill the 1/2-pint can with some of the mixed paint, stir, and return to the gallon of mixed paint. Stir until completely uniform.) Prepare a dry film of the mixed paint by using a film applicator (doctor blade) to draw down a film on a planar piece of opaque white glass at a wet film thickness of approximately 0.0076 centimeter. After a 48-hour air-drying period, compare the color of the film with the pertinent standard color card (see table III) in accordance with method 4250.1 of FED-STD-141. If doubt exists as to the satisfaction of the match, the color shall show not more than two color units difference from the standard when tested in accordance with method 6123 of FED-STD-141.

4.5 Toxicity. A contractor shall disclose the formulation of his product to NAVMEDCOM. The disclosure of proprietary information, which shall be held in confidence by NAVMEDCOM shall include: the name, formula, and approximate percentage by weight and volume of each ingredient in the product; the results of any toxicological testing of the product; identification of its pyrolysis products; and any such other information as may be needed to permit an accurate appraisal of any toxicity problem associated with the handling, storage, application, use, or disposal of the material.

4.6 Inspection of packaging. Sample packages and packs, and the inspection of preservation packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.



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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Packaging, packing and marking. The tinting colors shall be packaged level A or C, packed level A, B or C, and marked, as specified (see 6.2.1) in accordance with PPP-P-1892. Unless otherwise specified (see 6.2.1), the tinting colors shall be furnished in 1/2-pint cans.

5.1.1 Special marking. In addition to other markings required on the containers (see 6.2.1), there shall be the following statements: "The volatile content of the material in this container does not exceed 340 grams per liter."

"CAUTION: These colors contain toxic lead compounds. Do not use on interior surfaces, porches, or exterior surfaces of any items of furniture, machinery, or appliance to which children may be commonly exposed."

6. NOTES

6.1 Intended use. The tinting colors are intended for use in tinting enamel, nonflaming (dry), chlorinated alkyd resin, soft white semigloss formula no. 124 (of DOD-E-24607) to produce interior finish paints for living space on ships duplicating the enamels covered by DOD-E-24607. The tinting colors permit maintaining a wider selection of colors under conditions of limited storage space and inventory cost than possible with pretinted paints.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Tint color required (see 1.1).
(c) Levels of packaging and packing required (see 5.1).
(d) Size of container if other than specified (see 5.1).
(e) Special marking required (see 5.1.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraph.

| Paragraph no. | Data requirement title | Applicable DID no. | Option |
|---|---|---|---|
| 3.2 | Certificate of compliance | DI-E-2121 | ---- |



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(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Purchase unit. Tinting colors should be purchased by volume, the unit being a 1/2 U.S. pint at 15.6°C.

6.4 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets (MSDS) prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313. In order to obtain the MSDS, federal acquisition regulation (FAR) clause 52.223-3 must be in the contract.

6.5 Subject term (key word) listing.

Chlorinated alkyd
Mineral spirits
Thinner

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 8010-N368)



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